Started on	Friday, 29 May 2026, 11:27 PM
State	Finished
Completed on	Friday, 29 May 2026, 11:39 PM
Time taken	12 mins 7 secs
Grade	**100.00** out of 100.00

Question **1**

Correct

Mark 50.00 out of 50.00

Use LU Decomposition to find L and U matrix.

For example:

Input	Result
[[3, 2, 7], [2, 3, 1], [3, 4, 1]]	[[1. 0. 0.] [1. 1. 0.] [0.66666667 0.83333333 1.]] [[3. 2. 7.] [0. 2. -6.] [0. 0. 1.33333333]]

Answer: (penalty regime: 0 %)

Reset answer

```python
'''Program to find L and U matrix using LU decomposition.
Developed by:INIYA S
RegisterNumber: 212225230104
'''
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
from scipy.linalg import lu
matrix=np.array(eval(input()))
P,L,U=lu(matrix)
print(L)
print(U)
```

	Input	Expected	Got	
✓	[[3, 2, 7], [2, 3, 1], [3, 4, 1]]	[[1. 0. 0.] [1. 1. 0.] [0.66666667 0.83333333 1.]] [[3. 2. 7.] [0. 2. -6.] [0. 0. 1.33333333]]	[[1. 0. 0.] [1. 1. 0.] [0.66666667 0.83333333 1.]] [[3. 2. 7.] [0. 2. -6.] [0. 0. 1.33333333]]	✓

	Input	Expected	Got	

✓	`[[5, 1, 8], [4, 5, 7], [8, 9, 1]]`	`[[ 1.          0.          0.        ]`
		`[ 0.625       1.          0.        ]`
		`[ 0.5        -0.10810811  1.        ]]`
		`[[ 8.          9.          1.        ]`
		`[ 0.         -4.625       7.375     ]`
		`[ 0.          0.          7.2972973]]`
			`[[ 1.          0.          0.        ]`
			`[ 0.625       1.          0.        ]`
			`[ 0.5        -0.10810811  1.        ]]`
			`[[ 8.          9.          1.        ]`
			`[ 0.         -4.625       7.375     ]`
			`[ 0.          0.          7.2972973]]`
				✓

Passed all tests! ✓

▸ Show/hide question author's solution (Python3)

Correct

Marks for this submission: 50.00/50.00.

Question **2**

Correct

Mark 50.00 out of 50.00

Use LU Decomposition to solve a matrix.

For example:

Input	Result
`[[3, 2, 7], [2, 3, 1], [3, 4, 1]]` `[4, 5, 7]`	`[ 0.875  1.125 -0.125]`

Answer: (penalty regime: 0 %)

Reset answer

```
'''Program to solve a matrix using LU decomposition.
Developed by:INIYA S
RegisterNumber: 212225230104
'''
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
# To print X matrix (solution to the equations)
import numpy as np
from scipy.linalg import lu_factor,lu_solve
matrix=np.array(eval(input()))
constant=np.array(eval(input()))
piv,lu=lu_factor(matrix)
res=lu_solve((piv,lu),constant)
print(res)
```

	Input	Expected	Got	
✓	`[[3, 2, 7], [2, 3, 1], [3, 4, 1]]` `[4, 5, 7]`	`[ 0.875  1.125 -0.125]`	`[ 0.875  1.125 -0.125]`	✓

Passed all tests! ✓

▸ Show/hide question author's solution (Python3)

Correct

Marks for this submission: 50.00/50.00.